A polynomial error estimate for superpermutation length

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Abstract

A superpermutation on n letters is a word containing every permutation of those letters as a contiguous substring. Let $L(n)$ be its least possible length. We prove

$$L(n) = n! + (n-1)! + (n-2)! + 2(n-4)! + O\big((n-4)!n^{-1/61}\big).$$

The quantitative step uses a finite collection of joint label and cut states and a rational transfer argument. The latter gives the explicit coefficient $2 + 74/r$ once $n \geq 8 \cdot 10^{12} \cdot 352^4 r^{61}$, for $r \geq 6$ divisible by six. The accompanying Lean 4.35.0 formalization proves the finite covering-word bound and the two-sided asymptotic statement, including the construction of the source families and their assembly.

1 Words and the quantitative target

A *word* is a finite sequence of letters. A word on an alphabet of size n is a *superpermutation* if every permutation of the alphabet occurs in consecutive positions. For example, 121 is a superpermutation on two letters. Concatenating all permutations shows that a superpermutation exists for every positive n, so its minimum length $L(n)$ is defined. Put

$$F_3(n) = n! + (n-1)! + (n-2)!.$$

All factorials in this manuscript have nonnegative arguments.

The main finite bound uses the absolute integer $C = 8 \cdot 10^{12} \cdot 352^4$.

Theorem 1.1. *For every integer $r \geq 6$ divisible by six and every integer $n \geq Cr^{61}$, there is a superpermutation w on n letters such that*

$$r|w| \leq rF_3(n) + (2r + 74)(n-4)!. \tag{1}$$

We prove the theorem by constructing finite families of compatible rows and then assembling them into a word. The source families, their unused cyclic classes, and the auxiliary circles for assembly are all part of the construction. The final estimate is obtained by normalizing their exact integer counts.

Corollary 1.2. *As n tends to infinity through the positive integers,*

$$L(n) = F_3(n) + 2(n-4)! + O\Big((n-4)!n^{-1/61}\Big).$$

The matching lower estimate supplied by the formal development of [2] is

$$L(n) \geq F_3(n) + 2(n-4)! - 2(n-5)! \qquad (n \geq 7). \tag{2}$$

The precise imported declarations are `ConstAsym.lower2_L` and `ConstAsym.lower2_ratio`. The complete dependency chain of these results has been rebuilt under Lean 4.35.0. We use (2) only for the final two-sided consequence. Theorem 1.1 is established in Corollary 5.7. Selecting the largest admissible multiple of six then proves Corollary 1.2, as detailed in Corollary 5.8.

We first describe the literal local moves and the restricted collection of states that contains every permitted modification. Counting and separating those states gives many mutually disjoint local constructions. The next step combines constructions on overlapping sets of letters and removes their possible collisions. The assembly section describes the transport, the additional circles used for joining, and the extension by an arbitrary number of letters. Finally, a rational estimate converts the exact counts into (1), and an integer choice of r gives the exponent $1/61$.

2 Literal contours and finite sets of states

This section constructs many families of cyclic bases that remain disjoint after certain local changes. The local changes will later prevent intersections between families built on different sets of control symbols. The construction keeps track of the actual order of the symbols. This is needed because the word assembly uses literal overlaps, not just cyclic equivalence classes.

Fix an integer $r \geq 6$, and write $k = r-1$ and $d = r-3$. There are M distinct *background symbols*, in a fixed cyclic order, and r distinct *control symbols*. These two alphabets are disjoint. Number the background symbols by the elements of $\mathbb{Z}/M\mathbb{Z}$, in their cyclic order. A *cut* is the position immediately before a background symbol; its coordinate is the number of that symbol. There are k stationary positions for control symbols and one mobile position. The stationary positions have fixed identities $0, \ldots, k-1$. Their identities do not change when their cuts pass the distinguished background symbol.

A *literal base* is a list containing all $M + r$ symbols exactly once. Two bases have the same *cyclic class* if one is a cyclic rotation of the other. We always write the mobile symbol last. After deleting it, the remaining list is a representative of the stationary cyclic order. Thus the stationary cuts and their resident control symbols, together with the gap occupied by the mobile symbol, determine the literal base uniquely. The gap is necessary: the stationary cuts and the assignment of control symbols alone do not determine a base during a sequence of full moves.

Put $h = M + k$, so a base has length $h + 1$. For a base $b = b_0 \cdots b_h$, define two literal operations by

$$F(b) = b_1 \cdots b_{h-1} b_0 b_h, \qquad S(b) = b_h b_0 \cdots b_{h-3} b_{h-1} b_{h-2}.$$

A *full row* consists of a base with visible length $h + 1$, and a *short row* consists of a base with visible length $h - 1$. Their *charges* are respectively 0 and 2. A fixed extra symbol, called the satellite, is outside the base alphabet and is attached to every row when the rows are used in the word construction. It does not change in this section.

The head of a row is the first $h - 1$ symbols of its base. Its tail is the first $h - 1$ symbols after rotating the base left by one more than its visible length. Two consecutive rows are compatible when the first tail equals the second head. The displayed formulas show directly that the successor of a full row with base b may have base $F(b)$, and the successor of a short row may have base $S(b)$. A *closed contour* is a nonempty cyclic list of compatible rows. Its source classes are the cyclic classes of its bases, with one occurrence for each row.

2.1 Positive updates and local simplicity

Suppose a stationary position contains A, the background symbol at its cut is x, and the mobile symbol is T. Apply F until the base ends in A, x, T, and then apply S. We call this a *positive update* of that stationary position. The cyclic order changes locally from

$$\cdots A, x, T \cdots \quad \text{to} \quad \cdots x, A, T \cdots .$$

Consequently the old mobile symbol T becomes the stationary resident at the next background cut, and A becomes mobile. The new mobile is immediately before the new resident. In terms of the assignment of control symbols to positions, the update exchanges the mobile position with the chosen stationary position. No other stationary cut changes.

A *batch* updates every stationary position once, in some prescribed order. If the initial cuts are $q_0, \ldots, q_{k-1}$, then during batch t the current cuts are

$$q_i + t + \epsilon_i \pmod{M}, \qquad \epsilon_i \in \{0, 1\}. \tag{3}$$

Here ϵ_i is one exactly when position i has already been updated in the current batch. After the last update, increase t by one and reset all the ϵ_i to zero. The short row used for that last update still has the old value of t and has phase zero at its last target. Only its successor uses the reset convention. Full moves change the mobile gap and leave the data in (3) unchanged.

The circular distance between two residues modulo M is the smaller of the two distances between them around the M cuts. Call the initial cuts *robust* if the indexed family

$$0, \qquad q_i - q_j \quad (0 \le i, j < k,\ i \ne j)$$

has circular distance at least five between any two different members. The word "indexed" matters: repeated differences are forbidden. In particular, the cuts are distinct. Perturbing each cut by zero or one leaves distinct stationary cuts at circular distance at least four.

For example, with $r = 6$ and $M = 360$, the cuts

$$(q_0, q_1, q_2, q_3, q_4) = (0, 5, 25, 125, 155)$$

are robust. Their positive integer differences are $5, 20, 25, 30, 100, 120, 125, 130, 150, 155$. These differences, their negatives modulo 360, and zero are separated by at least five.

Lemma 2.1. *Assume the cuts are robust. After a positive update, the next positive update reaches its required suffix in fewer than $h - 1$ full moves.*

Proof. The stationary list has h distinct symbols. Each full move rotates it one position, so the required target suffix is reached in a unique number u of moves with $0 \le u < h$. It remains to exclude $u = h - 1$.

Let x be the background symbol crossed by the preceding short move. Immediately after that move, the stationary list ends in x, and the position just updated has cut $x + 1$. If $u = h - 1$, its final symbol x becomes its first symbol. On the other hand, let z be the cut of the next target. At the required suffix, the stationary list ends in that target's resident followed by z. The successor cut $z + 1$ is unoccupied, because the current stationary cuts are separated by at least four. Therefore the first stationary symbol at this suffix is the background symbol $z + 1$. We obtain $x = z + 1$ modulo M.

The previous target now has cut $x + 1 = z + 2$, whereas the next target has cut z. If these are different stationary positions, their distance two contradicts the separation just stated. If they are the same position, the equality would give $2 = 0$ modulo M, which is impossible: robust cuts include at least two different cuts at circular distance at least five, so $M \ge 10$. Thus $u \ne h - 1$, proving the stronger bound. □

Lemma 2.2. *Assume the cuts are robust. Consider a schedule of exactly M batches that returns the complete literal base to its initial value. Its source classes are pairwise distinct.*

Proof. Suppose two emitted rows have the same cyclic base. If their mobile symbols are equal, they have k common stationary control symbols; otherwise they have $k-1$. In either case there are at least two. A common stationary symbol occupies some position i in the first row and some position j in the second. Equality of its background cut gives

$$t' - t = q_i - q_j + \epsilon_i - \epsilon'_j \pmod{M}.$$

The ordered pairs (i, j) for different common symbols are distinct. Two off-diagonal pairs would place distinct nonzero ordered differences within distance two. One diagonal and one off-diagonal pair would place a nonzero difference within distance two of zero. Robustness excludes both possibilities. Therefore every common symbol occupies the same stationary position in both rows, and $t' - t$ is 0, 1, or -1 modulo M.

Choose the direction of comparison so that the rows lie in the same batch or in consecutive batches of the cyclic schedule. Literal closure permits this comparison across the last and first batches. In the same batch, equality of the phases of a common stationary symbol implies that its position is not updated between the rows. In consecutive batches, its phase must be one in the first row and zero in the second. It has already been updated before the first row, and is not updated again before the second. Thus no common stationary position is updated in the interval between the rows.

If the mobile symbols are equal, every stationary position is common. The two rows lie in a single run of full moves, possibly including its final short source row. The h possible insertion gaps for the mobile in the fixed stationary circle give h different cyclic bases. By Lemma 2.1, that run has fewer than h full moves, so no two of its source rows have the same cyclic base.

If the mobile symbols differ, write them as a and b. The only stationary position not common to the two rows is one position j, occupied by b first and by a second. All intervening updates must be at j. There is at most one such update in one batch, and at most two across the boundary between consecutive batches. The mobile changes, so the number of these exchanges is odd. It is therefore exactly one.

Let x be the cut of j before that update. Before the short move, b is stationary at x and a is mobile. Afterwards, a is stationary at $x+1$ and b is mobile. Equality of the two cyclic bases requires a to occupy cut $x+1$ before the update. There is only one such mobile gap, because no stationary resident is at that cut before the update. This is the gap of the short source row. The same equality requires the new mobile b to occupy cut x afterwards. Immediately after the short move, the stationary list begins with a and ends with the background symbol x. Inserting b before x therefore requires the final full gap, reached after exactly $h-1$ full moves. Lemma 2.1 says that the next target is reached strictly before this gap. The required row therefore does not occur. This excludes the final possibility. □

2.2 The normal order and exact replacement intervals

Choose representatives $0 = q_0 < q_1 < \cdots < q_{k-1} < M$. The *normal order* is

$$O = (k-1, k-2, \ldots, 0).$$

Begin with the mobile immediately before the resident at position zero. As before, the mobile is the last symbol of the literal base. A complete normal batch acts on the r control positions as an r-cycle, denoted C. Indeed, the original mobile goes to the first target; each subsequent target receives the resident displaced from the preceding target; and the resident of the last target becomes mobile.

Lemma 2.3. *If r divides M, then M normal batches form a literally closed contour. Its number of rows N_0 and its charge Q_0 are*

$$N_0 = M\big((r-2)M + r(r-1) + 1\big), \qquad Q_0 = 2(r-1)M.$$

Proof. The role assignment returns because C^M is the identity. Every stationary cut advances M times and therefore returns. The final target is again zero, so the mobile is again immediately before its resident. These data determine the same literal base, with the mobile last.

In one normal batch the first full run crosses $q_{k-1} + 1$ background symbols and all k resident controls. After updating target i, the full run to target $i - 1$ crosses $M + q_{i-1} - q_i$ background symbols and all k resident controls. Consequently the total number of full moves is

$$q_{k-1} + 1 + k + \sum_{i=1}^{k-1} (M + q_{i-1} - q_i + k) = (k-1)M + 1 + k^2.$$

There are also k short rows, each of charge two. Multiplying the two batch totals by M gives the formulas. □

We next describe a replacement that keeps a specified control symbol out of a specified stationary position for a chosen interval. Let P be an order of the k stationary positions, with first target a. Performing P followed by its reverse is a *parking pair*. Each position is updated twice. The exchanges of roles cancel in reverse order, so every control returns to its original role. The control that was initially mobile is placed at a by the first exchange and recovered by the last. No intermediate exchange uses a. Throughout the pair, including all full rows, this control is either mobile or resident at a.

For example, if $k = 5$ and the forbidden position is 4, use $P = (3, 4, 2, 1, 0)$. A control initially mobile stays out of position 4 throughout this order and its reverse. At the midpoint it is resident at position 3, although all stationary cuts have already advanced once.

Lemma 2.4. *Let T be a positive integer. Replace $2rT + 1$ consecutive normal batches by rT parking pairs, all with the same order P, followed by one normal batch. The replacement has exactly the same final literal base as the normal interval and the same total number of short rows. A control initially mobile never occupies any stationary position different from the first target of P during the parking portion.*

Proof. The parking pairs restore the role assignment, and the final normal batch applies C. The normal interval also applies C, because $C^{2rT+1} = C$. Both intervals advance every stationary cut $2rT + 1$ times. Both end with an update of position zero. The post-update mobile gap is therefore the same, and the literal bases agree. Each interval has $k(2rT + 1)$ short updates. The final assertion follows from the preceding description of each pair. Protection is asserted during the parking portion; the final normal batch must lie outside the interval to be protected. □

We call the final normal batch the *cleanup batch*. It is needed because a parking pair can end with its mobile immediately before a resident other than that of position zero. Equality of role assignments alone would not give equality of literal endpoints.

2.3 A catalogue of joint role and phase states

Only two parking orders are needed. Put $P_0 = O$, and obtain P_1 by exchanging the first two entries of O. For a forbidden position j, use P_1 when j is the first target of P_0, and use P_0 otherwise.

The chosen first target always differs from j. Fix this choice for all parking pairs in a replacement interval.

An *admissible schedule* is obtained from the normal M-batch contour by replacing mutually disjoint intervals as in Lemma 2.4, using only P_0 and P_1. An interval may be placed on the cyclic schedule; choose the initial cut outside the replacement intervals. Since each replacement has its normal literal endpoint, all replacements can be performed without changing the states at the other normal boundaries.

Let a role assignment be a bijection from the r positions, including the mobile position, to the r named controls. Write $U_\ell(P)$ for the role action of the first ℓ updates of P followed by its reverse, where $0 \le \ell \le 2k$. For such a prefix define a phase vector by

$$\epsilon_\ell(P) = \begin{cases} \text{the indicator of the first } \ell \text{ targets of } P, & 0 \le \ell < k, \\ \text{the indicator of the first } \ell - k \text{ targets of } P^{\mathrm{rev}}, & k \le \ell < 2k, \\ 0, & \ell = 2k. \end{cases}$$

At the midpoint $\ell = k$, the translation has increased by one and the phase vector is zero. The role action remains $U_k(P)$; it has not reset. At $\ell = 2k$, the translation has increased by two and both the phase vector and the role action have reset.

Define the finite set of *joint states*

$$\mathcal{J} = \left\{ \left(C^s U_\ell(P), \epsilon_\ell(P)\right) : 0 \le s < r,\ P \in \{P_0, P_1\},\ 0 \le \ell \le 2k \right\}. \tag{4}$$

Composition here means chronological action on a role assignment: the normal batches act first, followed by the indicated prefix. We count joint states, not independent choices of a role permutation and a phase vector.

Lemma 2.5. *Fix an initial role assignment σ. Every source row of an admissible schedule has stationary cuts $q_i + t + \epsilon_i$ and role assignment $\sigma\tau$ for some translation t and some $(\tau, \epsilon) \in \mathcal{J}$. Moreover,*

$$|\mathcal{J}| \le 2r(2k+1) = 4r^2 - 2r < 4r^2.$$

Proof. A normal boundary has assignment σC^s for some s modulo r. The proper prefixes of its batch are first-half prefixes of the P_0 pair. Its final state is the empty prefix at normal phase $s+1$.

A replacement starts at a normal boundary. Every complete parking pair restores its starting role assignment. Thus every pair in the replacement has the same list of relative role actions $C^s U_\ell(P)$. The reset convention in (3) gives exactly the displayed phase vectors, including the midpoint. The cleanup has the proper prefixes of a P_0 batch, and its final state is the next normal role phase. By Lemma 2.4, the subsequent normal boundary agrees with the original schedule, so the argument applies successively to all replacements.

Full moves change neither the role assignment nor the stationary cuts. The short row preceding the last update of a batch is represented by the proper prefix before that update; its successor is represented at the reset boundary. Thus every emitted row has been included. Finally, r divides M, the translation increases by M, and the schedule closes literally. The last and first states obey the same convention. The cardinality bound counts the indices in (4); repeated representations only decrease the number of states. □

2.4 Many robust modular frames

The stationary cut set will have one additional algebraic property that allows us to reconstruct a missing cut. Choose integers

$$0 < z_1 < \cdots < z_d < M, \qquad s = z_1 + \cdots + z_d \pmod{M},$$

where s is represented in $\{0, \dots, M-1\}$. Sort the $k = d+2$ points $0, z_1, \dots, z_d, s$ into increasing order, and retain the result only when it is robust. This ordered cut tuple is a *modular frame*. The point s is its sum point. Assign coefficient $d-1$ to zero, coefficient 1 to the sum point, and coefficient -1 to every z_i. These coefficients are all nonzero, their sum is zero, and their weighted sum of cuts is zero modulo M. Thus the relation is unchanged by a common translation. In the example above, $z = (5, 25, 125)$ and $s = 155$.

We use the following elementary fact for arbitrary moduli. For a nonzero integer a, the congruence $ax = b \pmod M$ has either no solutions or $\gcd(|a|, M)$ solutions in $\mathbb{Z}/M\mathbb{Z}$. Indeed, if one solution exists then all solutions form a translate of the kernel of multiplication by a. On writing $g = \gcd(|a|, M)$ and dividing by g, coprimality shows that the kernel consists of the g multiples of M/g. In particular, there are at most $|a|$ solutions. No primality assumption is needed.

Lemma 2.6. *Let F_M be the number of robust modular frames. Then*

$$F_M \geq \frac{(M-1)^d - 18k^4(M-1)^{d-1}}{d!} \geq \frac{M^d}{d!}\left(1 - \frac{19r^4}{M}\right).$$

Each retained anchored cut set has a unique sum point.

Proof. First choose an ordered vector from $\{1, \dots, M-1\}^d$, without sorting its coordinates. Regard the cuts as evaluations of the formal points $0, e_1, \dots, e_d, \mathbf{1}$ in $\mathbb{Z}^d$. Their affine relations form a one-dimensional space, generated by

$$(d-1, -1, \dots, -1, 1).$$

To verify this, a relation with coefficient a at $\mathbf{1}$ has coefficient $-a$ at each e_i, by comparing coordinates, and coefficient $(d-1)a$ at zero, by summing coefficients. Every nonzero relation therefore involves all $d+2 \geq 5$ points.

A comparison of two distinct ordered differences, or of a nonzero difference with zero, involves at most four formal points. Its linear form is therefore nonzero. Every coefficient in that form has absolute value at most two. There are at most k^4 comparisons. For robustness, exclude the nine residues $-4, -3, \dots, 4$ for each comparison. Fix all but one coordinate with nonzero coefficient. The modular root bound gives at most two choices for the remaining coordinate for each excluded residue. The number of bad ordered vectors is at most $18k^4(M-1)^{d-1}$.

The good vectors are invariant under coordinate permutation, and a good vector has distinct coordinates. Its orbit under coordinate permutation therefore has exactly $d!$ elements. Sorting gives the first inequality. For the second, use $(M-1)^d \geq M^d - dM^{d-1}$ and $d + 18k^4 \leq 19r^4$.

It remains to check that sorting has not identified different sum decorations. If an anchored cut set had two sum points s and s', the sum of all its cuts would be both $2s$ and $2s'$ modulo M. Distinct solutions require M even and $s - s' = M/2$. The two ordered differences $s - s'$ and $s' - s$ would then coincide. This contradicts robustness, so the sum point is unique. □

2.5 Reconstruction and disjoint labelled families

A *labelled frame* is a pair (q, σ), where q is a robust modular frame and σ is an initial bijective assignment of the r named controls to the r positions. Its *envelope* is the set of cyclic bases obtained by choosing an arbitrary translation, a joint state $(\tau, \epsilon) \in \mathcal{J}$, and an arbitrary mobile insertion gap, with stationary cuts $q_i + t + \epsilon_i$ and assignment $\sigma\tau$. Lemma 2.5 shows that every source row of every admissible repair of this labelled frame lies in its envelope.

Lemma 2.7. *Fix a labelled frame and fix a joint state on each side of a possible envelope intersection. The number of positional reconstruction witnesses for a target frame is at most*

$$D_0 = k(k-1)^2(d+1+\gcd(d-1,M)) \le 2d\,k(k-1)^2 \le 2r^4.$$

A witness includes the matching of the retained stationary positions.

Proof. A common cyclic base has at least $k-1$ controls stationary on both sides. If the mobile symbols agree, discard one of the k common stationary controls. Choose the omitted stationary position on the source side and on the target side. The remaining $k-1$ stationary controls occur in the same cyclic order. The binary phase perturbations do not change their order, because the cuts remain separated by at least four. There are therefore $k-1$ possible cyclic matchings.

For a matched source position i and target position j, equality of the cuts gives

$$q'_j + \delta = q_i + \epsilon_i - \epsilon'_j \pmod{M},$$

where δ is the difference of the two common translations. The phase vectors are fixed, so the right-hand side is known. Choose also the target sum position $a \in \{1,\dots,k-1\}$. This fixes its coefficient vector c. Set $w_j = q'_j + \delta$. Since $\sum c_j = 0$, the target relation is $\sum c_j w_j = 0$. It determines the omitted coordinate w_j in at most $\gcd(|c_j|, M)$ ways. If position zero was retained, its known coordinate w_0 determines δ; if it was omitted, its reconstructed coordinate does so. Subtracting δ then determines every target cut. Sorting and robustness can reject a reconstruction but cannot add one.

For each fixed sum position, summing the root multiplicities over the omitted target position gives $d+1+\gcd(d-1,M) \le 2d$. There are k source omissions, $k-1$ cyclic matchings, and $k-1$ choices of sum position. Multiplying gives the stated bound, which counts witnesses even if several reconstruct the same frame. □

Lemma 2.8. *Put*

$$\Delta = 64r^8, \qquad \eta = \frac{r!}{2d!(\Delta+1)}.$$

For every $M \ge 38r^4$, there are $\lfloor \eta M^d \rfloor$ labelled frames whose envelopes are pairwise disjoint.

Proof. Form a finite graph whose vertices are all labelled frames, with an edge when two distinct vertices have intersecting envelopes. Distinct vertices with coincident envelopes are also adjacent. By Lemma 2.6, the graph has at least $r!M^d/(2d!)$ vertices.

Fix a source vertex. For a chosen pair of joint states, each witness of Lemma 2.7 determines the target frame and matches $r-2$ named stationary control symbols. Since the target role action is a permutation, those equalities determine the target initial assignment on $r-2$ positions. The remaining two labels have at most two assignments to the remaining positions. Thus the number of possible neighbouring vertices, including the source vertex itself, is at most

$$2|\mathcal{J}|^2 D_0 \le 2(4r^2)^2(2r^4) = 64r^8.$$

This bound includes all assignments at the time the graph is formed. It would not be valid to select unlabelled frames first and then attach all $r!$ assignments to each selected frame.

Greedily select a vertex and remove it and all its neighbours. Each selection removes at most $\Delta+1$ vertices. Continuing until no vertices remain produces an independent set with at least $r!M^d/(2d!(\Delta+1)) = \eta M^d$ vertices. Select the required integer number from this independent set. Their envelopes are disjoint by the definition of the graph. □

For fixed M, the selected family can be used for every background circle and every set of r active letters. To see this, first make the selection using M formal background symbols and r different formal control symbols. Replace the background symbols, in order, by the letters of any chosen background circle, and replace the formal controls by its active letters. These substitutions give an injective map on the entire formal alphabet. They commute with the construction of each row and with cyclic rotation. If two substituted rows were equal up to rotation, injectivity would give the same equality before substitution. Thus the selected envelopes remain disjoint. In particular, the choice of frames is independent of the guard positions introduced in the next section.

2.6 What the local construction supplies to assembly

For later use, the literal displacement also has an exact description. Give each background symbol its index modulo h in a literal base. A full move changes that index by -1. A short move changes it by $+1$, except for the crossed background symbol, which stays at index $h-1$. In M complete batches each stationary position crosses every background symbol exactly once. Every background symbol therefore has exactly k exceptional short moves. If the contour is literally closed, with N rows and total charge Q, this proves

$$Q - N \equiv k \pmod{h}, \qquad \gcd(|Q-N|, h) = \gcd(k, h) \le k. \tag{5}$$

An untouched normal batch contains an actual consecutive run of literal F successors of length at least

$$\frac{(k-1)M + k^2 + 1}{k}.$$

It is an untouched batch that gives this estimate. The first full run of a cleanup batch can start at a different mobile gap and is not covered by the normal count. The global repair argument must therefore leave an untouched batch before applying a word assembly lemma that requires such a run.

The results in this section construct disjoint envelopes for one active control alphabet on one fixed background. They also prove that every admissible repair stays inside its envelope. To use several overlapping active alphabets, one must additionally construct the forbidden-event catalogue, prove the deletion estimates, and place disjoint replacement intervals. An actual word bound further requires the lists of unused source classes, an iteration that inserts a variable number of controls, and additional cyclic words that meet every required literal overlap. Those are separate mathematical statements; the numerical counts in this section do not assume their existence.

3 Combining the local constructions

Fix an integer $r \ge 6$ divisible by six, and put $k = r-1$ and $d = r-3$. We use the labelled frame family of the preceding section, with

$$\Delta = 64r^8, \qquad \eta = \frac{r!}{2d!(\Delta+1)}, \qquad \beta = (r-2)\eta.$$

At each sufficiently large background size M, select exactly $\lfloor \eta M^d \rfloor$ labelled frames whose allowed sets of cyclic rows are pairwise disjoint. The selection is fixed before the modifications described below. Define

$$A = \left\lceil \frac{r^3}{\beta} \right\rceil, \quad Q_{\mathrm{grid}} = rA, \quad W = r^2 A, \quad H = rA^2, \quad G = W - r, \quad \rho = \frac{A\beta}{r}, \quad E = 10^{12} r^{22} W^4. \tag{6}$$

The letter Q_{grid} denotes a grid width. It is unrelated to the row charge used in the assembly section.

3.1 Sets of active letters and fixed background letters

Take the W control letters to be the pairs (i, j) with $0 \le i < r$ and $j \in \mathbb{Z}/Q_{\text{grid}}\mathbb{Z}$. For $0 \le a < A$ and $b \in \mathbb{Z}/Q_{\text{grid}}\mathbb{Z}$, set

$$T_{a,b} = \{(i, ai + b) : 0 \le i < r\}.$$

The members of $T_{a,b}$ are the *active letters* of that construction. The other G control letters are its *guards*; these letters stay in the background circle and are never chosen as mobile letters. For example, when $r = 3$ and $A = 2$, the blocks $T_{0,0}$ and $T_{1,0}$ in a grid with six columns share just $(0, 0)$.

There are H blocks, each with r letters. Distinct blocks intersect in at most one letter. Indeed, two common letters in distinct rows i, j would imply

$$Q_{\text{grid}} \mid (a - a')(i - j), \qquad |(a - a')(i - j)| \le (A - 1)(r - 1) < Q_{\text{grid}}.$$

The product must therefore be zero. Since $i \ne j$, this gives $a = a'$, and either common letter then gives $b = b'$. This argument uses integers and remains valid when Q_{grid} is composite. Each letter belongs to A blocks. It follows that a block meets exactly $I = r(A - 1) \le W/r$ other blocks: each of its letters lies in $A - 1$ other blocks, and no block is counted at two letters.

Fix a cyclic order of m old letters. Insert each guard in a different old cut, immediately before the old letter at that cut. A *guard background* is the resulting cyclic order of $M = m + G$ letters. There are exactly

$$(m)_G = m(m - 1) \cdots (m - G + 1)$$

guard backgrounds for each block. Distinct cuts give no ordering multiplicity. For $m \ge G$,

$$1 - \frac{G(G - 1)}{2m} \le \frac{(m)_G}{m^G} \le 1. \tag{7}$$

The lower bound follows by applying $\prod_i (1 - u_i) \ge 1 - \sum_i u_i$ to $u_i = i/m$.

A *candidate* consists of an active block, a guard background, and one selected labelled frame. Candidates on the same block but different backgrounds cannot share a row: deleting their active letters would recover the same background from that row. Candidates on the same background are already separated by the selected frame family. Thus only candidates on different blocks can collide.

3.2 A finite list of necessary collision equations

Anchor every cyclic background at the same old letter. A local background cut $y \in \{0, \dots, M - 1\}$ projects to an original cut $x \in \{0, \dots, m - 1\}$. Counting the guards passed before y gives

$$y \equiv x + e \pmod{m}, \qquad 0 \le e \le G.$$

This also covers the guard immediately before the anchor. In that case the local cut has the form $m + e$ and the original cut is zero.

Every stationary frame satisfies a relation whose coefficient multiset is

$$(d - 1, 1, \underbrace{-1, \dots, -1}_{d \text{ entries}}).$$

The coefficients sum to zero, and their absolute values sum to $2d$. Including the binary cut phases and reducing the local relation modulo m shows that any foreign row must satisfy

$$\sum_u c_u x_u \equiv b \pmod{m}, \qquad |b| \le C_1 = (4d + 1)G + 2d \le 4rW. \tag{8}$$

For completeness, choose the canonical local cuts in $[0, M)$. Their weighted sum differs from the binary phase sum by an integer multiple of M, whose multiplier has absolute value at most $2d+1$. Replacing M by $m+G$ and each local cut by x_u+e_u gives the bound $(2d+1)G+2dG+2d$ displayed above. There are $k(k-1)$ assignments of this coefficient multiset to the stationary physical labels, since one chooses the label with coefficient $d-1$ and then the label with coefficient 1.

First delete a candidate if a relation (8) holds for a possible foreign stationary set containing none of its active letters. Such a relation has a nonzero coefficient on a guard variable. There are at most

$$J_1 \le Hr\,k(k-1)(2C_1+1) \le 9rW^3$$

relations to consider. A nonzero coefficient has absolute value at most r. After all other guard variables are fixed, its congruence has at most r solutions modulo m, because the number of roots of $ax=b$ is at most $\gcd(a,m) \le |a|$. Thus each equation excludes at most rm^{G-1} guard tuples. If $m \ge G^2$, (7) gives $(m)_G \ge m^G/2$. This first deletion removes at most the fraction

$$\frac{18r^2W^3}{m}. \tag{9}$$

The selected frame count cancels from this calculation.

Suppose two surviving candidates on different blocks share a cyclic row. Their blocks must have a common letter v, and v must be stationary in both rows. If the blocks were disjoint, all foreign stationary letters would be guards. If v were mobile in one row, its stationary letters would all be guards for the other candidate. Either case contradicts the first deletion, applied to the appropriate candidate.

If v occupies stationary slot j during batch t, then its local cut is $(q_j+t+\epsilon_j) \bmod M$. Here $0 \le t < M$ is the canonical batch number. The local wrap is zero or one, so its original cut obeys

$$x_v \equiv q_j + t + e' \pmod m, \qquad -2G \le e' \le 1.$$

Substituting in (8) gives the following finite list of possible events:

$$a(t+q_j) + \sum_u b_u x_u \equiv b_1 \pmod m, \qquad |b_1| \le C_2 = (6d-1)G + 3d - 1 \le 6rW. \tag{10}$$

An *event* forbids the pair (v, j) for that entire batch. The list ranges over every meeting foreign block, every foreign mobile letter, every coefficient assignment, every candidate slot j, and every integer b_1 in the stated interval. It therefore includes both the original rows and every permitted later modification.

There are at most

$$J_{\text{event}} \le Ik^3(k-1)(2C_2+1) \le 13r^4W^2$$

equation types. For a fixed type there are at most $2|a|$ actual roots $0 \le t < M$, provided $M < 2m$. For a fixed shared label, summing $|a|$ over its coefficient assignments gives $2d(d+1)$. Hence the total number of events, with repetitions allowed, is at most

$$K \le 4Ik^2d(d+1)(2C_2+1) \le 52r^4W^2. \tag{11}$$

3.3 Separating events without a distribution assumption

Put $L = 2r+1$ and $S = 2L$. Delete a candidate if it has two events for different forbidden pairs at circular distance at most S on the M-batch circle. Eliminating the two event times gives

$$aa'(q_j - q_l) + \sum_u (a'b_u - ab'_u)x_u \equiv a'b_1 - ab'_1 + aa'(\delta + \sigma G) \pmod m, \tag{12}$$

where $|\delta| \le S$ and $\sigma \in \{-1, 0, 1\}$. The term σG accounts for a wrap on the M-circle. It must be retained even though the congruence is modulo m. There are at most

$$J_2 \le 3(2S+1)J_{\text{event}}^2 \le 4563 r^9 W^4$$

such equations.

Every equation (12) has a nonzero variable coefficient. Here are all possibilities. If the foreign blocks differ, each has $r-2$ guard letters in its stationary set, and the two blocks share at most one letter. A private guard then has a nonzero coefficient. If the foreign block is the same but its mobile letters differ, a guard present in only one stationary set has a nonzero coefficient. If the block and mobile agree, either a guard coefficient $a'b_u - ab'_u$ is nonzero or the two full coefficient vectors are proportional. In the latter case their equal multisets force the ratio to be 1. Indeed their squared norms force the ratio to be 1 or -1, and $-(d-1)$ does not occur in the multiset when $d \ge 3$. Thus the vectors are equal. The forbidden pairs then differ only in their slots, so the remaining form is $a^2(q_j - q_l)$ with $j \ne l$. On a fixed position of the sum point, the cuts have the distinct coefficient vectors $0, e_1, \dots, e_d, \mathbf{1}$. Their difference is nonzero. All coefficients have absolute value at most $3r^2$.

We count the guard-variable and guard-free equations separately. If a guard coefficient is nonzero, fix the selected labelled frame and all other guards. There are at most $3r^2$ roots modulo m. Dividing by $(m)_G \ge m^G/2$ bounds the deleted fraction by $6r^2/m$ per equation, uniformly in the selected frame.

If all guard coefficients vanish, partition the sorted frame parameters according to the position of their sum point and the integer

$$\ell = \left\lfloor \frac{\sum_i z_i}{M} \right\rfloor.$$

There are at most $d(k-1) \le r^2$ choices. Within such a part, $q_j = v_j \cdot z - \mathbf{1}_{j=a}\ell M$. The winding term changes only the constant in (12). Fix a coordinate with nonzero coefficient. It ranges in an interval shorter than $2m$, so it has at most $6r^2$ roots. Deleting that coordinate leaves a strictly increasing list of $d-1$ integers, with at most $M^{d-1}/(d-1)!$ possibilities. Including all $r!$ initial assignments and dividing by $\lfloor \eta M^d \rfloor \ge \eta M^d/2$ bounds the deleted fraction, per part and per equation, by

$$\frac{12 r^2 r!}{\eta (d-1)! M} = \frac{24 r^2 d(\Delta+1)}{M}.$$

This counts in the entire parameter set before restricting to the selected frames. It requires no uniform distribution property of that selection. Since $\Delta + 1 \le 65 r^8$ and $M \ge m$, both cases together are bounded by

$$\frac{7\,118\,280 r^{22} W^4}{m}. \tag{13}$$

3.4 Modifying every surviving candidate

Around every event take the closed interval of radius L on the batch circle, and merge intersecting intervals. Each connected component has one forbidden pair, since intervals for different pairs cannot intersect. If there are no events, keep the normal construction. Otherwise, $M > (2L+1)K$ ensures a gap longer than $2L+1$ between consecutive event times. Cut in this gap and regard the remaining components as ordinary intervals. The cut changes only where these intervals are written; the canonical times in (10) remain unchanged.

For a component whose first and last events are f, e, let (v, j) be its forbidden pair. Choose the normal boundary $s \in [f - r + 1, f]$ at which v is mobile; the normal role action is a cycle on the r

roles. Set

$$P = 2r \left\lceil \frac{e+1-s}{2r} \right\rceil .$$

Replace the next P normal batches by $P/2$ pairs of a permitted order and its reverse, using the order whose first target is different from j. Follow them by one normal cleanup batch. The local endpoint theorem shows that these $P+1$ batches end at exactly the normal literal state. Throughout the first P batches, the letter v is either mobile or in the first target slot. Thus it never occupies j during any event batch.

The inequalities

$$f - r + 1 \le s \le f, \qquad e + 1 \le s + P \le e + 2r, \qquad s + P + 1 \le e + 2r + 1$$

put the entire replacement inside its merged interval. Different replacements are disjoint. Each restores its normal endpoint, so they can all be performed on the same cyclic walk. The joint-state containment proved earlier keeps every modified row inside the originally selected allowed set. If two modified candidates met, their unique shared letter would be stationary in both rows and would produce an event (10). The modification prevents that occupancy. This excludes collisions between two modifications as well as collisions with normal rows.

Regard a batch starting at integer time t as the unit interval $[t, t+1]$. The inequalities above put every changed batch inside the interval union. That union has total length at most $2LK \le 5rK$, so at most $5rK$ batches change. A complete batch has at most $k(M+k)$ source rows. Comparing each changed batch with its normal version therefore changes the total row count by at most $10r^2KM$ when $M \ge r$. The short charge stays exactly $2(r-1)M$. There is also an untouched normal batch. Its full rows follow the literal successor F, which is the additional hypothesis needed in the connector construction.

3.5 Uniform counts

Let B be the total number of source rows in the surviving modified contours, let q be their total short charge, and let c_{adv} be the number of circles in the actual connector lists constructed in the assembly section. Put $D_{m,W} = \prod_{i=0}^{W-1}(m+i)$ and $c_r = 2(r-1)/(r-2)$.

Proposition 3.1. *For every multiple m of r with $m \ge 4E$, the construction has literal closure, pairwise distinct source cyclic classes above the chosen old cyclic order, and the safe connector lists of the assembly section. Its counts satisfy*

$$\frac{B}{D_{m,W}} = \frac{\rho W}{m}(1+\theta), \qquad \frac{q + c_{\mathrm{adv}}}{D_{m,W}} = \frac{c_r \rho W}{m^2}(1+\epsilon), \qquad |\theta|, |\epsilon| \le \frac{E}{m},$$

and $c_{\mathrm{adv}} \le (r-1)q/m$.

Proof. We first verify the size conditions used in the construction. The ceiling defining A gives

$$\eta^{-1} = \frac{r-2}{\beta} \le \frac{r}{\beta} \le \frac{W}{r^4} \le W.$$

Also $W \ge r^2 \ge 1$ and $G \le W$. The definition of E implies $E \ge W^2$, $E \ge rW$, and $4E \ge 38r^4$. Hence

$$M \ge m \ge 38r^4, \qquad m \ge 4W^2 \ge G^2, \qquad G < m, \qquad M = m + G < 2m.$$

Since $d \geq 3$ and $M \geq m \geq 4W$, we have $\eta M^d \geq M/W \geq 4$. Moreover, $rG/m \leq rW/(4E) \leq 1/4$. Finally, the event bound gives

$$(4r+3)K \leq 260r^5W^2 < 4 \cdot 10^{12} r^{22} W^4 \leq m \leq M.$$

The strict middle inequality follows from $r, W \geq 1$ and $260 < 4 \cdot 10^{12}$. These prove all the required size conditions. Also $r \mid M$, since r divides m and W. The local closure and distinctness results and the preceding modifications consequently apply. Every source class is above the chosen old cyclic order. Deleting all controls first removes the guards from the guard background. A full move only rotates the remaining old cyclic order. A positive short update moves a control past one background letter, so it does not change the relative cyclic order of the old letters. This remains true during every replacement. Deleting all controls from any resulting row therefore recovers the chosen old class, as required by Lemma 4.4.

Each modified contour has

$$N = (r-2)M^2 + [r(r-1)+1]M + \delta_N, \quad |\delta_N| \leq 10r^2KM, \quad q_0 = 2(r-1)M.$$

Let $h = M + r - 1$ be the source port modulus and let g be its greatest common divisor with the monodromy. The local result gives $1 \leq g \leq r-1$. Write $k = r-1$. The untouched normal batch has k short rows and $(r-2)M + r(r-1) + 1$ rows in total. One of its k consecutive full runs therefore has length

$$R \geq \frac{(k-1)M + k^2 + 1}{k}.$$

Its bases follow the literal successor F, and the local construction gives $R \leq h$. To check the numerical hypothesis of the connector lemma, use $g\lfloor R/g \rfloor \geq R - g$ and $g \leq k$. These give

$$2g\lfloor R/g \rfloor \geq 2R - 2g \geq \frac{2(k-1)M + 2}{k} \geq M + k = h.$$

For the last inequality, it suffices that $(k-2)M + 2 \geq k^2$. This follows from $M \geq 2k$ and $k \geq 5$, since $2k(k-2) + 2 \geq k^2$. Thus the connector lemma supplies an actual list of at most $2g^2 \leq 2(r-1)^2$ circles for each contour. Summing over the contours gives $c_{\mathrm{adv}} \leq (r-1)q/M \leq (r-1)q/m$.

Before deletion the number of candidates is $C_0 = H(m)_G \lfloor \eta M^d \rfloor$. Define

$$\phi = \frac{(m)_G}{m^G}, \qquad \psi = \frac{\lfloor \eta M^d \rfloor}{\eta M^d}, \qquad \mu = \frac{M}{m}, \qquad \chi = \frac{D_{m,W}}{m^W}.$$

The exact leading contributions before deletion are

$$C_0(r-2)M^2 = H\beta m^{W-1}\phi\psi\mu^{d+2},$$
$$C_0 q_0 = c_r H\beta m^{W-2}\phi\psi\mu^{d+1}.$$

The exponents follow from $G + d + 2 = W - 1$ and $G + d + 1 = W - 2$. Let t be the proportion of candidates that survive both deletions. The deletion estimates give

$$1 - \frac{18r^2W^3 + 7\,118\,280r^{22}W^4}{m} \leq t \leq 1.$$

In particular $t > 0$ under the present size bound. Averaging the relative row errors over the surviving contours gives a number u such that

$$|u| \leq \frac{2r + 1040r^5W^2}{m}.$$

Their short charges are all exactly q_0. Write $v = c_{\mathrm{adv}}/q$, so $0 \le v \le (r-1)/m$. The two normalized counts now have the exact expressions

$$\frac{B}{D_{m,W}} = \frac{H\beta}{m}\frac{\phi\psi t\mu^{d+2}(1+u)}{\chi},$$
$$\frac{q+c_{\mathrm{adv}}}{D_{m,W}} = \frac{c_r H\beta}{m^2}\frac{\phi\psi t\mu^{d+1}(1+v)}{\chi}.$$

The relative-error numerators, each to be divided by m, are bounded by

rounding the selected frame count	W,
distinct guard positions	$W^2/2$,
guard-only deletion	$18r^2W^3$,
nearby incompatible events	$7\,118\,280r^{22}W^4$,
row count within a contour	$2r + 1040r^5W^2$,
additional connector charge	r,
replacing powers of M by powers of m	$2W^2$,
the product $D_{m,W}/m^W$	W^2.

The first bound uses $d \ge 3$ and $\eta^{-1} \le W/r^4$. The next three were proved in (7), (9), and (13). The row bound follows by dividing the displayed estimate for N by $(r-2)M^2$. We use the product inequalities

$$\prod_i (1+u_i) \le \frac{1}{1-\sum_i u_i}, \qquad \prod_i (1-u_i) \ge 1 - \sum_i u_i \quad (0 \le u_i,\ \sum_i u_i < 1).$$

For the first inequality, use $1+u_i \le 1/(1-u_i)$ and the second inequality. For the powers of μ, there are at most r equal factors $1+G/m$. Their deviations sum to at most $rG/m \le 1/4$, so

$$1 \le \mu^{d+1} \le \mu^{d+2} \le \frac{1}{1-rG/m} \le 1 + \frac{2rG}{m} \le 1 + \frac{2W^2}{m}.$$

For χ, the deviations i/m sum to at most $W^2/(2m) \le 1/8$, which gives $1 \le \chi \le 1 + W^2/m$ in the same way.

The sum of the listed numerators is less than $9 \cdot 10^6 r^{22} W^4$. Let s be that sum divided by m. Then $s < 1/1000$. To combine the estimates, let s_0 be the sum of the deviations allowed for the numerator factors in either exact expression, and let s_1 be the deviation allowed for χ. Thus $s_0 + s_1 \le s$. The product inequalities give a numerator between $1 - s_0$ and $1/(1-s_0)$, while $1 \le \chi \le 1 + s_1$. The ratio is therefore at least

$$\frac{1-s_0}{1+s_1} \ge 1 - s_0 - s_1 \ge 1 - s$$

and at most $1/(1-s_0) \le 1/(1-s) \le 1 + 2s$. In particular its deviation from one is at most $4s$. Since $36 \cdot 10^6 < 10^{12}$, each relative error is at most E/m. Finally $H\beta = \rho W$, as required. □

Lemma 3.2. *The parameters in* (6) *satisfy* $r^2 \le W \le 352r^9$ *and* $r^2 \le \rho < r^2 + 1$.

Proof. Cancellation of factorials gives

$$\beta = \frac{r(r-1)(r-2)^2}{2(64r^8+1)}.$$

For $r \geq 6$, the inequalities $r - 1 \geq 5r/6$ and $r - 2 \geq 2r/3$ imply $\beta \geq 1/(351r^4)$. Also $0 < \beta < 1$. Thus

$$r^2 \leq W = r^2 A < \frac{r^5}{\beta} + r^2 \leq 351r^9 + r^2 \leq 352r^9.$$

The defining ceiling for A gives $r^3 \leq A\beta < r^3 + \beta$. Division by r yields the bounds on ρ. □

4 From source contours to covering words

The numerical recurrence used later concerns finite lists of rows and circles. We describe those lists and prove that an extension may use any positive number of new letters. The one-letter transport lemmas in the supplied assembly library apply repeatedly to give this construction.

4.1 Rows, closed contours, and insertion

Write $\mathrm{rot}(b, j)$ for the word obtained by rotating a nonempty word b to the left through j positions. A *cyclic class* consists of all rotations of a word with distinct letters. If x is absent from b, write $\mathrm{ins}(b, x, j)$ for insertion of x immediately before position j, where positions start at zero and position $|b|$ means insertion at the end. Thus $\mathrm{ins}(abc, x, 1) = axbc$ and $\mathrm{ins}(abc, x, 3) = abcx$.

A *row* is a triple (b, s, v), where b has distinct letters, the letter s does not occur in b, and $1 \leq v \leq |b|$. The word b is its *base*, the letter s is its *satellite*, and its *charge* is $|b| - v$. Its head and tail are the following literal words:

$$\mathrm{head}(b, s, v) = b[0 : |b| - 2], \qquad \mathrm{tail}(b, s, v) = \mathrm{rot}(b, v + 1)[0 : |b| - 2].$$

Here $b[i : j]$ means the consecutive letters in positions i through $j - 1$. A row is *full* if its charge is zero, and *short* if its charge is two. Except during the final completion, all rows below are full or short. All rows in a given contour have the same satellite and the same base alphabet.

Two rows are *compatible* if the first tail equals the second head as a word. A *closed contour* is a nonempty finite list of rows in which each row is compatible with the next, including the last and first rows. Repeated occurrences remain separate entries of the list. For example, the full rows with bases *abcd*, *bcad*, and *cabd*, all with one fixed satellite outside $\{a, b, c, d\}$, form a closed contour. Their heads are ab, bc, ca, respectively, and their tails are bc, ca, ab.

For a base of length $h + 1$, an *insertion port* is an integer in $\{0, \ldots, h - 1\}$. These are the h insertion positions in a head of length $h - 1$. Across a full row the port changes from p to $p - 1$ modulo h, and across a short row it changes from p to $p + 1$ modulo h. Consequently the port displacement of a row is its charge minus one. If a contour has N rows and total charge q, its displacement is $\Delta = q - N$.

We recall the literal transport operation from the supplied assembly library [3]. Let z be a fresh ordinary letter. A full row $(b, s, h + 1)$ at port zero is replaced by the two full rows with bases $\mathrm{ins}(b, z, 0)$ and $\mathrm{ins}(b, z, h)$. At a port $1 \leq p < h$ it is replaced by the single full row with base $\mathrm{ins}(b, z, p)$. A short row at a port $p < h - 1$ is replaced by the single short row with base $\mathrm{ins}(b, z, p)$. At port $h - 1$ it is replaced by two short rows with bases $\mathrm{ins}(b, z, h - 1)$ and $\mathrm{rot}(b, h)z$. The satellite is unchanged in all four cases. A transported contour is formed by concatenating these local paths and propagating the port between successive source rows.

Each local path starts at the old head with z inserted at the input port and ends at the old tail with z inserted at the output port. This follows directly from the displayed definitions of head and tail. For example, when $b = aUb$ is a full base and the input port is zero, the two new bases are

$zaUb$ and $aUzb$. Their common interface is the literal word aU. These endpoint equations explain why compatible source rows give compatible transported paths.

Across all h input ports, each source row gives exactly $h+1$ new row occurrences. All those occurrences have its charge. Their base cyclic classes are precisely the classes obtained by inserting z in all cyclic gaps of the source base. The endpoint, inventory, and coverage statements are the local transport propositions imported from the supplied library [3]. The formulas above specify the operation to which those propositions apply.

Lemma 4.1. *Let a closed contour have bases of length $h+1$, where $h \geq 2$, and put $g = \gcd(h, |\Delta|)$ and $\ell = h/g$. For each $0 \leq e < g$, transport ℓ consecutive copies of the contour from initial port e. The resulting g contours are closed. Their displacements are divisible by $h+1$, and their combined row and charge counts are $(h+1)N$ and $(h+1)q$, respectively. Every source-row and source-port pair occurs once in their transport.*

Proof. The ports at the beginnings of the successive source laps in component e are $e + t\Delta$ modulo h, for $0 \leq t < \ell$. Consider the map

$$(e, t) \longmapsto e + t\Delta \pmod{h}, \qquad 0 \leq e < g, \quad 0 \leq t < h/g.$$

Equality of two outputs first gives equality of e modulo g, and thus equality of e in its specified range. Division of the remaining congruence by g then gives $(t - t')\Delta/g \equiv 0 \pmod{h/g}$. The integers Δ/g and h/g are coprime, so $t = t'$ in the specified range. The domain and codomain both have h elements, so the map is a bijection. If $\Delta = 0$, this argument reduces to $g = h$, $\ell = 1$, and the identity enumeration of the ports.

After ℓ laps the source port returns to e, since h divides $\ell\Delta$. The literal endpoint equations therefore close the transported path. The winding proposition for transport gives displacement $(h+1)\Delta/g$ for that path, which is divisible by $h+1$. This is the repeated-contour case of the imported winding theorem; its divisibility hypothesis is exactly $h \mid \ell\Delta$.

At any fixed source-row occurrence, the prefix port change is a translation of the set of lap-start ports. The bijection above therefore shows that all its input ports occur once across the g components. The local inventory now gives $h+1$ output rows, each with the source charge, for every source-row occurrence. Summing proves both count formulas. □

4.2 Explicit circles meeting every completion port

We reserve a second fresh letter f for final completion. For precision, we describe the completion operation used in the imported joining theorem. For a row (b, s, v), its lift at position j has base $\mathrm{ins}(b, f, j)$, satellite s, and visible count $v + \mathbf{1}_{j<v}$. Its repair row at position j has base $\mathrm{rot}(b, j)s$, satellite f, and visible count $|b| + 1$. For a full row the completion path is its full transport path with new letter f. For a short row at port $p < h-1$ the path is its one lift at p. At port $h-1$ the path is, in order, the lift at $h-1$, the repair rows at $h-1$ and h, and the lift at h. As before, concatenate paths and propagate the source port. This defines the *completed contour* from each tagged initial port.

A *circle* is a word of distinct letters, considered together with all its rotations. Its displayed representative and its position in a list of circles are retained for counting. For a contour with bases of length $h+1$, the auxiliary circles have length h. We say that a circle *safely meets* a tagged completion port if either it contains f and one of its rotations is the head of a row in the completed contour, or the source contour is the full chart of a rotation of that circle. The *full chart* of a word d and a letter a outside d consists of the full rows with bases $\mathrm{rot}(d, j)a$, one for each $0 \leq j < |d|$.

This definition includes the circle d used when a pending class is closed. A list of circles is *safe* for a contour if every tagged initial completion port is safely met by at least one circle in the list.

One useful sufficient witness for the first kind of meeting consists of an actual source-row occurrence, the completion port propagated to that occurrence, and a rotation identity

$$\mathrm{rot}(C, t) = \mathrm{ins}(\mathrm{head}(b, s, v), f, p).$$

The first row of the local completion path has this head. Thus the identity witnesses an actual vertex of the completed contour. Both the occurrence and the propagated port matter; an equality of unlabeled cyclic gaps is insufficient.

Lemma 4.2. *Suppose a source path of port modulus h starts at port e, where $0 \le e < h$. If its displacement is S, the displacement of its transported path is*

$$S + \left\lfloor \frac{e + S}{h} \right\rfloor.$$

Immediately before the second transported row at a full source row whose input port is zero, the transported prefix displacement is congruent to $-e - 1$ modulo $h + 1$.

Proof. The formula is zero for the empty path. A full step subtracts one from S. The floor decreases by one precisely when the old residue of $e + S$ is zero. Thus the claimed expression decreases by two at port zero and by one at every other port. These are exactly the numbers of full rows in the two possible full transport paths. A short step adds one to S. The floor increases by one precisely at port $h - 1$, so the expression increases by two there and by one elsewhere. These are exactly the displacements of the short transport paths. Induction proves the formula. The floor is the Euclidean floor, also when $e + S$ is negative.

At a full port-zero occurrence, write $e + S = ht$. The first of its two transported rows is full. Just before the second row, the displacement is therefore $S + t - 1 = (h + 1)t - e - 1$, which gives the congruence. □

Lemma 4.3. *In the setting of Lemma 4.1, suppose the source contour has a consecutive run of R full rows with bases*

$$\mathrm{rot}(H, j)a, \qquad 0 \le j < R,$$

where $|H| = h$, a is absent from H, and $R \le h$. If $2g\lfloor R/g \rfloor \ge h$, the transported contours admit a safe list of at most $2g^2$ circles. Each circle is supported on the source alphabet together with f and omits the satellite.

Proof. Put $L = \lfloor R/g \rfloor$, and let S_0 be the displacement before the run begins. Fix a component with initial source port $e < g$, and let a_e be the residue of $e + S_0$ in $\{0, \dots, g - 1\}$. The full source rows at phases

$$j = a_e + gu, \qquad 0 \le u < L,$$

all lie in the run. Indeed, $a_e + g(L - 1) \le gL - 1 < R$. Each is reached at port zero in exactly one of the ℓ laps. To see this, its input port in lap t is $e + t\Delta + S_0 - j$ modulo h. The congruence modulo g holds by the choice of a_e, and after division by g the remaining congruence has one solution for t modulo ℓ.

Write $H_e = \mathrm{rot}(H, a_e)$. For each $0 \le t < g$, use the two circles

$$C_t^- = \mathrm{ins}(H_e, f, g(L - 1) + t), \qquad C_t^+ = \mathrm{ins}(H_e, f, h - g + t).$$

Their insertion positions lie between zero and $h-1$, because $gL \le R \le h$. Both circles have length $h+1$, contain f, and have distinct letters. At the second transported row of phase $a_e + gu$, the head is $\mathrm{rot}(H_e, gu)$. In particular the transport letter z is absent from this head.

For $0 \le j \le b < h$, splitting a word at positions j and b gives

$$\mathrm{rot}(\mathrm{ins}(H_e, f, b), j) = \mathrm{ins}(\mathrm{rot}(H_e, j), f, b-j),$$
$$\mathrm{rot}(\mathrm{ins}(H_e, f, b), b+1) = \mathrm{ins}(\mathrm{rot}(H_e, b), f, h).$$

The second identity is a separate endpoint identity. It is needed because the literal ports zero and h of the advanced contour are distinct.

Given a port $gv + t$ with $0 \le v < L$ and $0 \le t < g$, choose $u = L-1-v$ in the first identity for C_t^-. Thus the low circles meet local ports $0, \dots, gL-1$. The high circles, with u varying through $0, \dots, L-1$, meet local ports $h-gL, \dots, h-1$. For C_0^- at $u = L-1$, the second identity additionally meets port h. The hypothesis $2gL \ge h$ says that the first two intervals cover $0, \dots, h-1$.

Finally fix a tagged initial completion port J of this advanced component. Lemma 4.2 shows that its local port at every selected second row is $J - e - 1$ modulo $h+1$. Choose the local port representative in $\{0, \dots, h\}$ and use the corresponding circle and occurrence above. The resulting literal identity proves safe incidence from J. There are $2g$ circles for each of g components, so the total is $2g^2$. Coincident circles may remain as separate list occurrences. Every circle uses only H_e and f, which proves the support statement. □

The literal hypothesis on the full run is necessary. Compatibility alone allows a different full successor. For example, the three full bases $012, 120, 201$ form a compatible contour, but their first two-letter prefixes are not rotations of one fixed two-letter word. The positive-slot construction must supply a run from an untouched normal batch, where successive bases have precisely the form required in Lemma 4.3.

4.3 The finite universe above an old cyclic class

Fix a word d on $m \ge 2$ distinct old letters and distinct new letters $v_0, \dots, v_{W-1}$. A cyclic class on these letters is *above* d if deletion of all the v_i leaves the cyclic class of d. Define lists $U_i(d)$ recursively. Start with $U_0(d) = [d]$. For each b in $U_i(d)$, place

$$\mathrm{rot}(b, j)v_i, \qquad 0 \le j < m + i,$$

in $U_{i+1}(d)$, retaining the order of the parent list and the indicated order of j. For example, from $d = ab$ and one new letter x this gives $U_1(d) = [abx, bax]$.

Lemma 4.4. *The list $U_W(d)$ contains exactly one representative of each cyclic class above d. Its length is*

$$D_{m,W} = \prod_{i=0}^{W-1} (m+i).$$

If B source-row occurrences have pairwise distinct base cyclic classes above d, their complement in $U_W(d)$ is an actual list of length $D_{m,W} - B$.

Proof. We prove the first assertion by induction on i. It holds for $i = 0$. Given a class above d on the first $i+1$ new letters, delete v_i. The remaining class has a unique representative b in $U_i(d)$ by induction. Rotate the larger word so that its unique occurrence of v_i is last. The preceding word is a rotation of b, which gives existence of a representative in the displayed list.

For uniqueness, equality of two larger cyclic classes implies equality of their deleted parent classes. Thus their parent representatives agree. Because v_i occurs once and is last in both displayed representatives, cyclic equivalence forces equality of those representatives as literal words. A word of $m+i$ distinct letters has $m+i$ different rotations, so the two rotation indices agree as well. Each parent therefore contributes exactly $m+i$ new representatives. Multiplication over the steps gives the formula for $D_{m,W}$.

Each selected source occurrence has a unique representative in $U_W(d)$. The distinctness assumption makes these representatives distinct. Delete these B entries from the finite list. The resulting complement has length $D_{m,W} - B$, as asserted. This counting operation does not change the literal bases in the source contours. □

4.4 Extension by an arbitrary number of letters

We now specify the intermediate objects. Reserve distinct letters s and f for the satellite and final completion. A *stage of old size m* consists of an old alphabet A_0 of size m, a distinguished ordinary letter z outside A_0, a list of contours on $z \cup A_0$, a list of auxiliary circles of length m, and a list of pending cyclic classes on A_0. The alphabets omit s and f. Each contour has displacement divisible by m. Every circle has distinct letters and is supported on $\{f, z\} \cup A_0$, and the circle list is safe for every contour. Finally every cyclic class on $z \cup A_0$ either occurs as a row base or is obtained by inserting z into a pending cyclic class. Write R, Q, c, P for the row count, total charge, circle count, and pending count. Counts refer to list occurrences.

We use the following existing transport facts. Transporting every component from every port through a fresh ordinary letter preserves closure, displacement divisibility and coverage. It multiplies row and charge counts by $m+1$. Extending each circle through all its m cyclic insertion positions gives mc circles, preserves distinct letters and support, and safely meets every transported component. These statements are the general one-letter level and circle transport results in the supplied assembly library [1]. They apply to arbitrary stages satisfying the preceding conditions.

Lemma 4.5. *Let a stage have old size $m \geq 2$, counts (R, Q, c, P), old alphabet A_0, and distinguished letter v_0. Let $W \geq 1$, and choose distinct fresh ordinary letters $v_1, \ldots, v_{W-1}, z'$. Suppose that, above every pending old class d, there is a family of closed source contours on $A_0 \cup \{v_0, \ldots, v_{W-1}\}$ with B pairwise distinct source classes above d and total charge q. Suppose its transport through z' has displacement divisible by $m+W$ and admits a safe supported list of c_{adv} circles of length $m+W$. Then there is a stage of old size $m+W$ and distinguished letter z' with counts*

$$\begin{aligned} R' &= A_{m,W} R + (m+W) P B, \\ Q' &= A_{m,W} Q + (m+W) P q, \\ c' &= D_{m,W} c + P c_{\mathrm{adv}}, \\ P' &= P(D_{m,W} - B), \end{aligned}$$

where $A_{m,W} = \prod_{i=1}^{W}(m+i)$ and $D_{m,W}$ is as in Lemma 4.4. The transport of the source family is required to include every source-row and source-port pair once.

Proof. Set $A_0' = A_0 \cup \{v_0, \ldots, v_{W-1}\}$. First transport the existing contours successively through $v_1, \ldots, v_{W-1}, z'$, and extend their circles through the same sequence. There are exactly W transports: the old distinguished letter v_0 already belongs to every existing component. At each step the letter being added is absent from the current ordinary alphabet and from both reserved roles. The one-letter results therefore apply inductively. They give closed contours on $z' \cup A_0'$, displacement divisible by $m+W$, valid supported circles, and safety from every completion port. They multiply

the existing row and charge counts by $(m+1)\cdots(m+W) = A_{m,W}$ and the circle count by $m\cdots(m+W-1) = D_{m,W}$.

For every old pending occurrence d, append its transported source contours and its supplied circles. Their structural hypotheses put them on the same alphabet with the same port modulus as the transported old components. Taking the union preserves closure and displacement divisibility component by component. It also preserves circle validity and support. A circle which met a component before taking the union is still in the union, so safety is preserved. Replace d in the pending list by the actual complement list from Lemma 4.4. Every new pending representative is a permutation of A_0'.

It remains to prove the coverage condition. Let y be an arbitrary cyclic class on $z' \cup A_0'$. Delete the new letters in the reverse order $z', v_{W-1}, \ldots, v_1$. This produces a cyclic class x on $v_0 \cup A_0$. Apply the old coverage condition to x.

If x is the class of an existing row, reinsert $v_1, \ldots, v_{W-1}, z'$ in the gaps determined by y. The one-letter coverage proposition at each transport gives a final transported row in class y.

Otherwise x is above an old pending class d. Delete z' from y and call the resulting source class t. Deleting all W source controls from t gives d, so t has one representative in $U_W(d)$. If that representative was selected, the source family has a row with base class t. The exhaustive transport inventory and the local insertion coverage supply a transported row in the class y. If it was not selected, its representative lies in the new pending complement, and y is obtained from that representative by inserting z'. These two alternatives establish the new coverage condition.

Each pending old occurrence contributes $(m+W)B$ advanced rows and charge $(m+W)q$, by the source-row and source-port inventory. It contributes exactly c_{adv} circles and $D_{m,W} - B$ pending classes. Adding these contributions to the transported old objects gives all four displayed counts. □

For the source family of Proposition 3.1, each surviving candidate contributes the closed contour obtained by its specified modifications. Order the finite set of surviving candidates and list these contours in that order. The local distinctness theorem and the exclusion of collisions between candidates show that all their source-row cyclic classes are distinct. Their row and charge counts are the sums of the counts of those same contours. In particular, selecting this list requires no additional choice of rows or cyclic classes.

Each contour has an untouched normal batch and thus a full run of the form required in Lemma 4.3. The verification in Proposition 3.1 gives the inequality for its length. Lemmas 4.1 and 4.3 consequently provide the transported contours, their complete source-port inventory, and their safe circle lists. Thus the family satisfies every source hypothesis of Lemma 4.5.

A family constructed on one fixed old order can be used above another old order by a single bijection of the full finite alphabet. Apply this bijection to the old letters, all source controls, the satellite, the transport letter, and the completion letter together. Rotation, insertion, prefixes, and equality of words commute with this relabeling. It therefore preserves literal compatibility and every incidence witness, as well as the counts. Relabeling only part of a frame would not justify those equalities.

4.5 An imported family for the final extensions

For the last part of the schedule, we use an existing family with three source controls. We state both its geometric properties and the numerical bounds that will be used. Put

$$\mathcal{G}_a = \{g \in \mathbb{Z} : 0 \le g < a, \quad g \equiv 1 \pmod 3, \quad 2g+5 \le a\}, \qquad \nu_a = \begin{cases} 2, & a \text{ odd}, \\ 8, & a \text{ even}. \end{cases}$$

Proposition 4.6. *For every old cyclic order on $a \geq 7$ letters, three distinct source controls, a fresh transport letter, and two further distinct reserved letters, there is a source family satisfying the hypotheses of Lemma 4.5 with width three. Its counts are*

$$B_a = 2|\mathcal{G}_a|a(a+12), \qquad q_a = 16|\mathcal{G}_a|a, \qquad c_{\mathrm{adv},a} = (a+2)\nu_a.$$

Writing $D_{a,3} = a(a+1)(a+2)$, these counts satisfy $B_a \leq D_{a,3}$ and $\nu_a \leq 8$. For $a \geq 49$, they also satisfy

$$D_{a,3} \leq 4B_a, \qquad a(q_a + c_{\mathrm{adv},a}) \leq 8B_a.$$

This is the family `UniR.rFam` in the supplied assembly library [1]. Its construction is valid for an arbitrary full frame of distinct letters satisfying the two reserved-letter exclusions. The declarations `rData_ok` and `rData_safe` give literal source validity, distinct source classes, and safe incidence from every required port. The constructor `Uniform.famS` supplies the transported components, their displacement divisibility, and their finite complement. Thus this proposition gives actual families and circle lists for Lemma 4.5.

The count formulas and the bounds $B_a \leq D_{a,3}$, $\nu_a \leq 8$, and $D_{a,3} \leq 4B_a$ are respectively `rowsR_eq`, `chargeR_eq`, `rowsR_le`, `ncycR_le`, and `rows_quarter` in `UniR/RRate.lean`. The cost inequality is `cost_le8` in `UniR/RRate8.lean`. The count of advanced circles is $(a+2)\nu_a$, since each of the original ν_a circles has $a+2$ extensions at transport. Whole-frame relabeling as described above transfers these statements to the chosen old cyclic order and ordinary letters. No parity restriction remains, and the family can extend an already existing stage. It need not start a new construction.

4.6 Initialization, padding, and the final word

At old size m, choose no contours and no circles, and take every cyclic class on the old alphabet to be pending. There are $(m-1)!$ such classes: fix one old letter first and permute the remaining $m-1$ letters. Every class on the old alphabet together with the distinguished letter lies above one of them, by deletion of that letter. Thus this is a stage with counts $(0, 0, 0, (m-1)!)$.

The identity

$$(m+W)D_{m,W} = mA_{m,W}$$

and Lemma 4.5 show that

$$R' + (m+W)P' = A_{m,W}(R + mP).$$

Consequently every stage constructed from this initialization satisfies $R + mP = m!$. Taking $W = 1$ and an empty selected family gives a valid one-letter extension with

$$(R', Q', c', P') = ((m+1)R, (m+1)Q, mc, mP).$$

Such extensions permit adjustment of the final alphabet size without a congruence restriction.

At a final old size h, replace each pending representative d by its full chart with the current distinguished letter, and add the circle d. The chart is closed, has h rows, has zero charge, and has displacement $-h$. It covers every class above d and is safely met by its own circle according to the definition. Thus the resulting closed level has $h!$ rows, charge Q, and $c + P$ circles, and covers every base cyclic class.

We use the following precise consequence of the imported word assembly theorem [1]. Suppose $h \geq 8$. A closed level on an ordinary alphabet of size $h+1$, with $h!$ rows, charge Q, and a valid supported safe list of C circles, gives a covering word on $n = h+3$ letters of length at most

$$F_3(n) + Q + hC + (a-2)C + (h+1-a)\min\{C, (n-1)^{\underline{h+1-a}}\}, \qquad 2 \leq a \leq h+1.$$

The notation $x^{\underline{k}}$ means $x(x-1)\cdots(x-k+1)$. The completed alphabet, consisting of the ordinary letters, satellite, and completion letter, is relabeled bijectively to $\{0,\dots,n-1\}$, with the satellite labeled 9 and the completion letter labeled 10. These labels are available because $n = h+3 \geq 11$. All other labels can be assigned bijectively to the ordinary letters. The relabeling preserves contiguous occurrences and meets the alphabet hypothesis of the imported declaration `Closed.word`. The theorem supplies an actual word containing every permutation, rather than an estimate for an abstract count.

Taking $a = 2$, $C = c + P$, and bounding the minimum by C gives

$$|w| \leq F_3(n) + Q + (2h-1)(c+P). \tag{14}$$

This is the completion estimate used in the finite transfer argument. The source families of Proposition 3.1, followed by the families of Proposition 4.6, maintain all of these hypotheses through the extensions above.

5 A finite transfer estimate

We now estimate the length supplied by the assembly construction. The starting size will be close to n, and most extensions will use the families of Proposition 3.1. A final sequence of extensions of size three reduces the pending fraction far enough for completion. We first give the transfer with explicit hypotheses and then verify them using the constructions in the preceding sections.

5.1 The construction statements needed for the estimate

For a stage of old size $m \geq 2$, retain the counts R, Q, c, P from Section 4. Define the rational numbers

$$y = \frac{Q}{m!}, \qquad z = \frac{c}{(m-1)!}, \qquad p = \frac{P}{(m-1)!}, \qquad X = y + z.$$

There are $(m-1)!$ cyclic classes on m letters, so p is the fraction of classes still awaiting an assignment. Normalizing the other counts by the indicated factorials will make inherited contributions unchanged under an extension.

An extension from size m to size $m+v$, where v is a positive integer, uses nonnegative integer counts B, q, c_{adv}. Put

$$D_{m,v} = \prod_{i=0}^{v-1}(m+i), \qquad A_{m,v} = \prod_{i=1}^{v}(m+i).$$

Lemma 4.5 gives the bounds and update rules

$$0 \leq B \leq D_{m,v}, \tag{15}$$

$$\begin{aligned} R' &= A_{m,v}R + (m+v)PB, \\ Q' &= A_{m,v}Q + (m+v)Pq, \\ c' &= D_{m,v}c + Pc_{\mathrm{adv}}, \\ P' &= P(D_{m,v} - B). \end{aligned} \tag{16}$$

The distinctness hypothesis in Lemma 4.4 is required for (15) and for the complement count in the last line.

Assume that the construction can start at any size $m_0 \geq 11$ with

$$(R, Q, c, P) = (0, 0, 0, (m_0-1)!).$$

Use also the empty extensions from Section 4, which have size one and $B = q = c_{\mathrm{adv}} = 0$. At the final intermediate size $h = n - 3$, bounding the minimum in (14) by $c + P$ gives an actual superpermutation w with

$$|w| \le F_3(n) + Q + (2h - 1)(c + P). \tag{17}$$

In the conditional results below, the initialization, the required extensions, and completion to an actual superpermutation are all hypotheses. Equations for counts without those construction statements are insufficient.

Lemma 5.1. *The rules* (16) *preserve the identity* $R + mP = m!$. *With*

$$f = \frac{B}{D_{m,v}}, \qquad g = \frac{q + c_{\mathrm{adv}}}{D_{m,v}},$$

their normalized form is

$$p' = p(1 - f), \qquad y' = y + p\frac{q}{D_{m,v}}, \qquad z' = z + p\frac{c_{\mathrm{adv}}}{D_{m,v}}, \qquad X' = X + pg. \tag{18}$$

In particular, $0 \le p \le 1$ *throughout the construction, and empty extensions preserve* y, z, p.

Proof. Cancellation in the products gives

$$(m + v)D_{m,v} = mA_{m,v}, \qquad (m + v)! = A_{m,v}m!, \qquad (m + v - 1)! = D_{m,v}(m - 1)!.$$

Consequently,

$$R' + (m + v)P' = A_{m,v}R + (m + v)PD_{m,v} = A_{m,v}(R + mP).$$

This is $(m + v)!$ when $R + mP = m!$. The initial counts satisfy that identity. Dividing the updates for Q', c', P' by the appropriate new factorials proves (18). Initially $p = 1$. The inequality $0 \le f \le 1$ therefore proves $0 \le p \le 1$ by induction. For an empty extension, $f = g = q = c_{\mathrm{adv}} = 0$, so the normalized quantities do not change. □

5.2 Two finite inequalities

The main estimate uses two short induction arguments. They apply to rational numbers and do not require an approximation to a product.

Lemma 5.2. *Let* $f_0, \dots, f_{J-1}$ *lie in* $[0, 1]$. *Set* $p_0 = 1$ *and* $p_{i+1} = p_i(1 - f_i)$. *Then*

$$p_J \left(1 + \sum_{i=0}^{J-1} f_i\right) \le 1.$$

Proof. Write $s_j = \sum_{i<j} f_i$. All the numbers p_j, s_j are nonnegative. Using $s_{j+1} = s_j + f_j$, we obtain

$$\begin{aligned} p_{j+1}(1 + s_{j+1}) &= p_j(1 - f_j)(1 + s_j + f_j) \\ &= p_j(1 + s_j) - p_j f_j(s_j + f_j) \\ &\le p_j(1 + s_j). \end{aligned}$$

The initial value is $p_0(1 + s_0) = 1$. Induction proves the claim. □

For example, if $f_0 = f_1 = 1/2$, then $p_2 = 1/4$, and the left side of the lemma is $1/2$. The estimate remains valid if a selected density is one, in which case all subsequent pending mass is zero.

Lemma 5.3. *Suppose $p_{i+1} = p_i(1 - f_i)$ and $X_{i+1} = X_i + p_i g_i$, with $p_i \geq 0$. If $t g_i \leq K f_i$ for every $i < J$, where $t > 0$, then*

$$t(X_J - X_0) \leq K(p_0 - p_J).$$

Proof. For each i,

$$t(X_{i+1} - X_i) = t p_i g_i \leq K p_i f_i = K(p_i - p_{i+1}).$$

Summing cancels all intermediate X_i and p_i and gives the result. □

5.3 The schedule and its cost

Fix an integer $r \geq 6$ divisible by six. Let W be a positive multiple of r, let $E > 0$, let $\rho \geq r^2$, and put

$$c_r = \frac{2(r-1)}{r-2}.$$

At every multiple m of r with $m \geq 4E$, assume that the construction provides an extension of size W for which

$$f_m = \frac{\rho W}{m}(1 + \theta_m), \qquad g_m = \frac{c_r \rho W}{m^2}(1 + \varepsilon_m), \qquad |\theta_m|, |\varepsilon_m| \leq \frac{E}{m}, \tag{19}$$

and

$$c_{\mathrm{adv},m} \leq \frac{r-1}{m} q_m. \tag{20}$$

The final extensions come from Proposition 4.6. For an extension of size three at any $a \geq 49$, its inequalities and Lemma 5.1 give

$$p' \leq \frac{3}{4}p, \qquad X' - X \leq \frac{8}{a}(p - p'), \qquad z' - z \leq \frac{8p}{a(a+1)}. \tag{21}$$

The extensions of size W select a small proportion of the available classes. The extensions of size three remove at least one quarter of the pending mass. We use the latter only at the end.

Here is the derivation of (21). Its selected fraction is $f = B_a/D_{a,3} \geq 1/4$, so $p' = p(1-f) \leq 3p/4$. The cost estimate of Proposition 4.6 gives

$$X' - X = p\frac{q_a + c_{\mathrm{adv},a}}{D_{a,3}} \leq \frac{8}{a}\frac{pB_a}{D_{a,3}} = \frac{8}{a}(p - p').$$

Finally, its circle count gives

$$z' - z = p\frac{(a+2)\nu_a}{a(a+1)(a+2)} = \frac{p\nu_a}{a(a+1)} \leq \frac{8p}{a(a+1)}.$$

These are updates of the existing stage. In particular, the final extensions retain the rows and safe circles obtained during the sparse part of the construction.

For a positive integer n, let ℓ be the least nonnegative integer such that $n < 2^\ell$. Define

$$K = 12\ell, \qquad \tau = 3 + 36\ell.$$

For example, $\ell = 7$ when $n = 64$, because the upper inequality in the definition is strict.

Theorem 5.4. *Assume the construction and completion statements above. If*

$$n \geq 8rE, \qquad n \geq 96r^2, \qquad r + W + \tau \leq \frac{n}{2r}, \tag{22}$$

then there is a superpermutation w on n letters such that

$$|w| \leq F_3(n) + \left(2 + \frac{74}{r}\right)(n-4)!.$$

Proof. Put $\alpha = (r-1)/r$, and let m_0 be the least multiple of r that is at least αn. Thus

$$\alpha n \leq m_0 < \alpha n + r.$$

Set $a = n - \tau$ and

$$J = \left\lfloor \frac{a - m_0}{W} \right\rfloor, \qquad b = m_0 + JW.$$

Since $\lfloor x \rfloor > x - 1$, the last condition in (22) gives

$$J > \frac{n/r - r - \tau - W}{W} \geq \frac{n}{2rW} > 0.$$

In particular $J \geq 1$. Starting at m_0, use J extensions of size W, then $a - b$ empty extensions, and finally K extensions of size three. The floor defining J gives $0 \leq a - b < W$. The final size is $a + 3K = n - 3$, as required for completion.

The initial size satisfies $m_0 \geq n/2 \geq 48r^2 \geq 11$. Each start $m_i = m_0 + iW$ of an extension of size W is a multiple of r and lies between $n/2$ and n. Since $n \geq 8rE$, it is at least $4E$. Also $a \geq n/2$, because $\tau \leq n/(2r)$, and $a \geq n/2 \geq 48r^2 > 49$. Thus every extension used in this schedule is within its stated range. The last extension of size three starts at $n - 6$.

Write

$$\delta = \frac{2E}{n} \leq \frac{1}{4r}.$$

The two relative errors at each m_i have absolute value at most δ. In particular $f_{m_i} > 0$. From (19),

$$\frac{n g_{m_i}}{f_{m_i}} \leq \frac{c_r}{\alpha} \frac{1+\delta}{1-\delta}.$$

Here

$$\frac{c_r}{\alpha} = \frac{2r}{r-2} = 2 + \frac{4}{r-2} \leq 2 + \frac{6}{r},$$

because $r \geq 6$. Since $0 \leq \delta \leq 1/2$,

$$\frac{1+\delta}{1-\delta} = 1 + \frac{2\delta}{1-\delta} \leq 1 + 4\delta \leq 1 + \frac{1}{r}.$$

Finally,

$$\left(2 + \frac{6}{r}\right)\left(1 + \frac{1}{r}\right) = 2 + \frac{8}{r} + \frac{6}{r^2} \leq 2 + \frac{9}{r}.$$

Let p_s, X_s, y_s, z_s be the normalized values after the extensions of size W. Lemma 5.3, with $t = n$, now gives

$$nX_s \leq \left(2 + \frac{9}{r}\right)(1 - p_s) \leq 2 + \frac{9}{r}. \tag{23}$$

We next bound p_s. Each selected density satisfies

$$f_{m_i} \geq \frac{\rho W}{n}(1-\delta) \geq \frac{\rho W}{2n}.$$

The lower bound on J therefore yields

$$\sum_{i<J} f_{m_i} \geq \frac{n}{2rW}\frac{\rho W}{2n} = \frac{\rho}{4r} \geq \frac{r}{4}.$$

Lemma 5.2 proves

$$p_s \leq \frac{4}{r+4} \leq \frac{4}{r}. \tag{24}$$

The intervening empty extensions preserve these normalized quantities.

All the final extensions start at sizes at least a. Their cost bound in (21) telescopes to

$$X_{\text{end}} - X_s \leq \frac{8}{a}(p_s - p_{\text{end}}) \leq \frac{8p_s}{a}.$$

Using $n/a \leq 2$, (23), and (24), we obtain

$$nX_{\text{end}} \leq 2 + \frac{73}{r}. \tag{25}$$

The completion estimate also involves z_{end}. During the extensions of size W, (20) gives

$$z_s \leq \frac{r-1}{m_0} y_s.$$

Their local cost satisfies $m_i g_{m_i} \leq 8 f_{m_i}$, since $m_i \leq n$ and $2 + 9/r \leq 8$. Applying Lemma 5.3 with $t = m_0$ gives $y_s \leq X_s \leq 8/m_0$. Hence

$$z_s \leq \frac{8(r-1)}{m_0^2}.$$

For the final extensions, write p_j for the pending mass before the jth one. The contraction $p_{j+1} \leq 3p_j/4$ implies $p_j \leq 4(p_j - p_{j+1})$. Summing gives $\sum_{j<K} p_j \leq 4(p_s - p_{\text{end}}) \leq 4$. The last inequality in (21) consequently gives

$$z_{\text{end}} - z_s \leq \frac{32}{a(a+1)}.$$

Both m_0 and a are at least $n/2$, so

$$z_{\text{end}} \leq \frac{32r + 96}{n^2}. \tag{26}$$

There are 12ℓ final extensions. Since $p_s \leq 1$ and $(3/4)^4 = 81/256 < 1/2$, their contraction gives

$$p_{\text{end}} \leq (3/4)^{12\ell} < 2^{-3\ell} < n^{-3}. \tag{27}$$

The strict inequalities are valid because $\ell \geq 1$ and $n < 2^\ell$.

Put $h = n - 3$. Divide (17) by $(n-4)! = (h-1)!$. The normalized excess is at most

$$\begin{aligned} hy_{\text{end}} + (2h-1)(z_{\text{end}} + p_{\text{end}}) &= hX_{\text{end}} + (h-1)z_{\text{end}} + (2h-1)p_{\text{end}} \\ &\leq nX_{\text{end}} + nz_{\text{end}} + 2np_{\text{end}}. \end{aligned}$$

Equations (25)–(27) bound this by

$$2 + \frac{73}{r} + \frac{32r+96}{n} + \frac{2}{n^2}.$$

For $r \geq 6$, we have $32r + 96 \leq 48r$. The assumption $n \geq 96r^2$ therefore bounds the penultimate term by $1/(2r)$. It also implies $n^2 \geq 4r$, so the last term is at most $1/(2r)$. Their sum is at most $1/r$, which proves the theorem. □

5.4 One explicit threshold

Choose the parameters in (6). By Lemma 3.2, they satisfy $r^2 \le W \le 352r^9$ and $r^2 \le \rho < r^2 + 1$. We need one further consequence of those bounds.

Lemma 5.5. *The integer W is divisible by r, and $E \le 10^{12}352^4 r^{58}$.*

Proof. The identity $W = r^2 A$ proves divisibility. The upper bound for W gives

$$E = 10^{12} r^{22} W^4 \le 10^{12} r^{22} (352r^9)^4 = 10^{12} 352^4 r^{58}.$$

□

Lemma 5.6. *For these parameters, the condition $n \ge 8rE$ implies every condition in* (22).

Proof. Since $W \ge r^2 \ge 1$, we have $E \ge W$ and $E \ge 10^{12} r^{22} \ge 2592r$. Consequently

$$n \ge 8rE \ge (144r)^2.$$

This proves $n \ge 64$ and $n \ge 96r^2$.

Recall that ℓ is the least integer with $n < 2^\ell$. Thus $\ell \ge 7$ and $2^{\ell-1} \le n$. For every integer $t \ge 7$,

$$t^2 \le 2^{t-1}.$$

Indeed, the initial case is $49 \le 64$, and $(t+1)^2 \le 2t^2$ for $t \ge 7$, which proves the induction step. It follows that $\ell^2 \le n$. If $\ell \le 144r$, then $144r\ell \le (144r)^2 \le n$. If $\ell \ge 144r$, then $144r\ell \le \ell^2 \le n$. In either case,

$$36\ell \le \frac{n}{4r}.$$

Finally, $W \ge r^2 \ge r + 3$, and therefore

$$r + W + 3 \le 2W \le 2E \le \frac{n}{4r}.$$

Adding the last two inequalities gives $r + W + \tau \le n/(2r)$, as required. □

Corollary 5.7. *For every integer $r \ge 6$ divisible by six and every integer $n \ge Cr^{61}$, there is an actual superpermutation w satisfying*

$$r|w| \le rF_3(n) + (2r + 74)(n - 4)!.$$

Proof. Use the parameters of (6). At each multiple m of r with $m \ge 4E$, Proposition 3.1 supplies actual source families above any chosen old cyclic order. Its distinct source classes lie in the universe of Lemma 4.4, so $0 \le B \le D_{m,W}$ and the required complement exists. The same proposition supplies the literal full run needed by Lemma 4.3. Together with Lemma 4.1, this gives the safe supported circle lists and transported components required by Lemma 4.5.

Initialization and empty extensions are provided in Section 4. The final extensions and their normalized bounds are provided by Proposition 4.6 and (21). The schedule has final old size $n - 3$, and $n \ge Cr^{61} \ge 11$, so (14) gives its actual superpermutation. This verifies all the construction and completion hypotheses of Theorem 5.4.

Lemma 5.5 gives

$$8rE \le Cr^{59} \le Cr^{61} \le n.$$

Lemma 5.6 supplies the other size conditions. Apply Theorem 5.4 and multiply its conclusion by the positive integer r. □

5.5 The exponent in the asymptotic statement

Corollary 5.8.

$$L(n) = F_3(n) + 2(n-4)! + O\Big(\frac{(n-4)!}{n^{1/61}}\Big).$$

Proof. For $n \geq C6^{61}$, choose the largest positive multiple r of six with $Cr^{61} \leq n$. Such an r exists, and maximality gives

$$Cr^{61} \leq n < C(r+6)^{61} \leq C(2r)^{61}.$$

The last inequality uses $r \geq 6$. Taking positive sixty-first roots now gives

$$\frac{1}{r} < 2C^{1/61}n^{-1/61}.$$

The word from Corollary 5.7 yields

$$\frac{L(n) - F_3(n)}{(n-4)!} - 2 \leq \frac{74}{r} < 148C^{1/61}n^{-1/61}.$$

On the other side, use $(n-4)! = (n-4)(n-5)!$ to divide the imported lower bound (2) by $(n-4)!$. Since $n \geq 8$, this gives

$$2 - \frac{L(n) - F_3(n)}{(n-4)!} \leq \frac{2}{n-4} \leq \frac{4}{n} \leq 4n^{-1/61}.$$

Both sides are bounded by a constant times $n^{-1/61}$. Multiplying by $(n-4)!$ proves the stated remainder. □

6 Formal verification and scope

The accompanying formalization proves Theorem 1.1 and Corollary 1.2. It constructs the literal source rows, their repairs and safe connector circles, and the globally distinct source families. The exact counts of these objects supply the estimates used by the arbitrary-width extension. The rational transfer and the original word-assembly theorem then produce an actual covering word. Combining its length bound with the rebuilt lower-bound proof gives the two-sided result.

The final declarations are `finite_covering_word` and `shortest_length_asymptotic` in the namespace `CoefficientTwoPrecision`. They use the original coverage predicate and minimum-length definition. Neither theorem assumes a source family, a numerical recurrence, or the existence of a word with the stated bound. The accompanying proof map matches the mathematical steps to their Lean declarations.

The formalization uses Lean 4.35.0. Its pinned Mathlib revision is

`56a005855ea17e0b0958f73ec801ae140b0a1679`.

Build receipts record the checked source hashes, dependency versions, compiler results, and axiom reports. The final theorems depend only on Lean's `propext`, `Classical.choice`, and `Quot.sound`. The formal proof contains no admitted step or additional mathematical axiom. The manuscript and its component arguments have also received independent AI review; human mathematical review remains pending.

References

[1] Superpermutations: Supplied lean assembly source snapshot. Repository `https://github.com/jhalfar/superpermutations`, 2026. Revision 1a1fef9382bff3b557120ac535b4bda752397140. Directory proofs/lean/constant43586371; supplied with the project.

[2] Superpermutations: Supplied lean lower-bound and asymptotic sources. Repository `https://github.com/jhalfar/superpermutations`, 2026. Revision 1a1fef9382bff3b557120ac535b4bda752397140. Directory proofs/lean/asym/ConstAsym, especially Remainder8.lean; supplied with the project.

[3] Jay Pantone. Superpermutation upper bound: Supplied lean source snapshot. Repository `https://github.com/jaypantone/superperm-upper-43-80`, 2026. Revision c8fb7ffdd69be1375580710ae11b47a295b1f0f6. Literal rows, transport, and word assembly; supplied with the project.